



IILM UNIVERSITY, GREATER NOIDA

APPLIED AI + COMPUTER NETWORKING

NetSage AI

AI-Assisted Network Troubleshooting System with Human Review

PRESENTED BY

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ROLL NO.

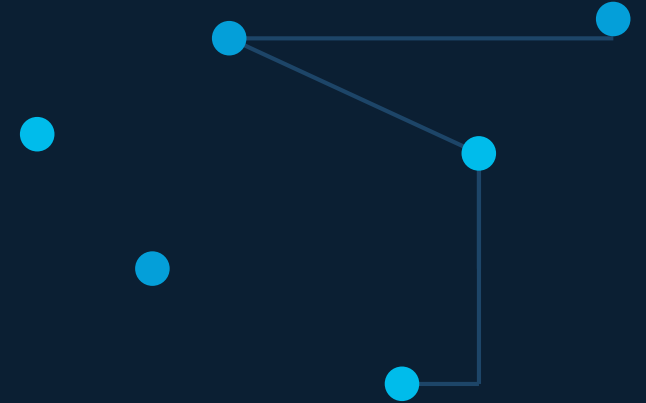
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UNIVERSITY

IILM University, Greater Noida

PROJECT DOMAIN

Applied AI + Computer Networking



Introduction

- Python-based network troubleshooting system
- Designed around Cisco-style troubleshooting scenarios
- Combines deterministic rules with an AI-oriented diagnosis architecture
- Produces structured troubleshooting recommendations
- Uses mandatory human review
- Follows a responsible AI approach

Rules

Deterministic Python checks on Cisco-style evidence

AI-Oriented Diagnosis

Structured, evidence-backed recommendation

Human Review

Engineer accepts, edits, or rejects the result

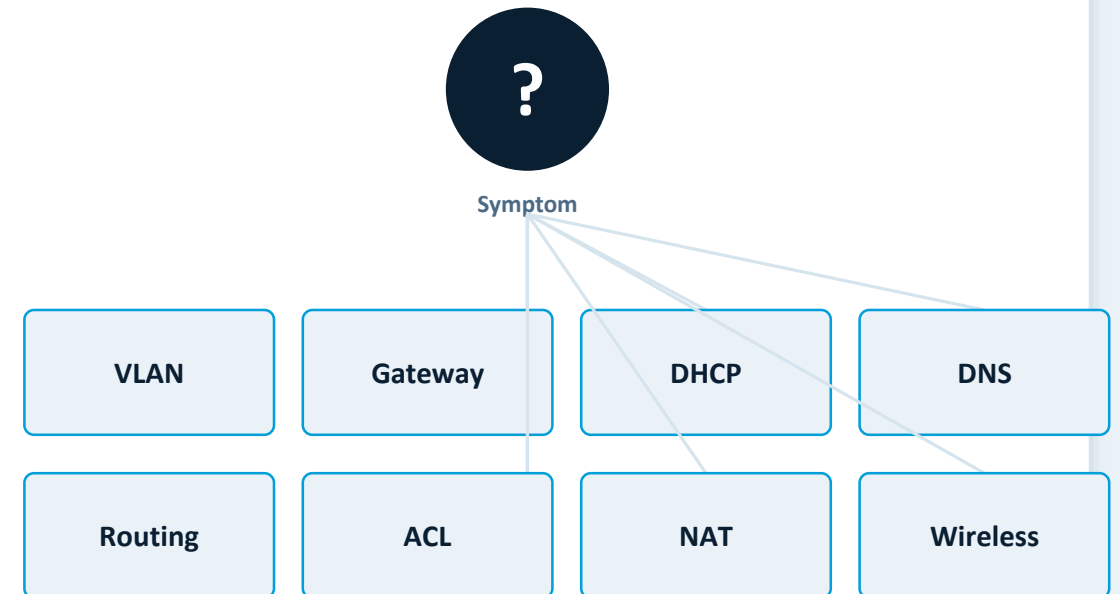
Why Troubleshooting Is Hard

- Network troubleshooting can be complex
- A single symptom can have multiple possible causes
- Junior engineers may struggle to identify root causes
- Manual diagnosis can be time-consuming
- Incorrect assumptions may result in incorrect fixes

"PC receives an IP address but cannot communicate with a server."

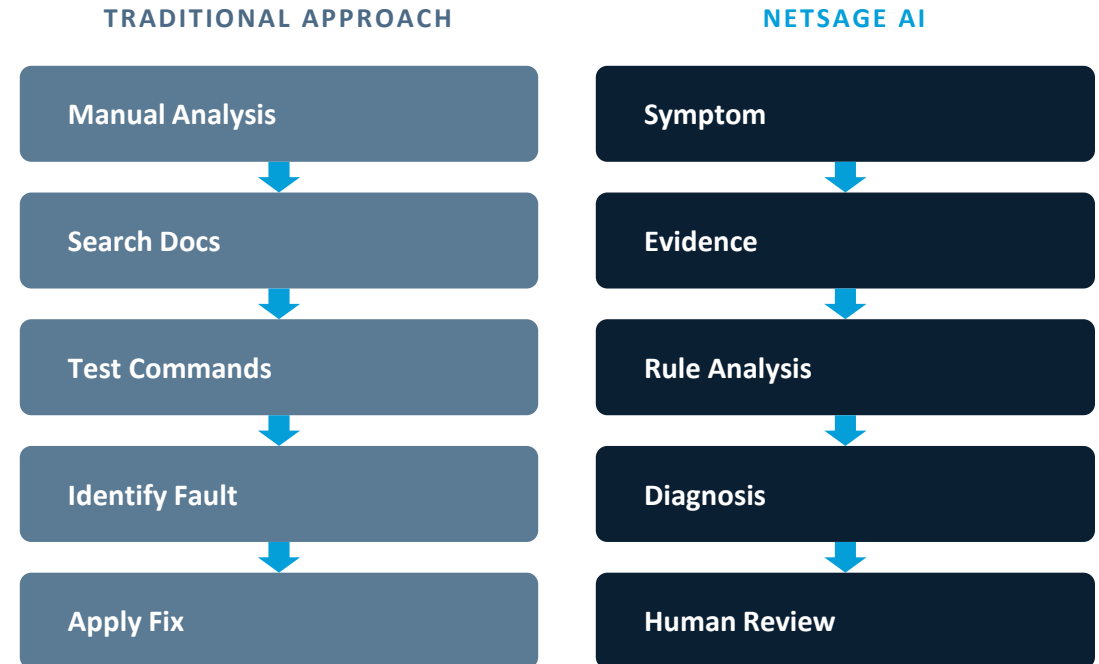
ONE SYMPTOM — MANY POSSIBLE CAUSES

ONE SYMPTOM, MULTIPLE FAULTS



Why an Intelligent Assistant Helps

- Faster troubleshooting assistance
- Structured diagnostic process
- Better support for junior engineers
- Evidence-based recommendations
- Reduced guesswork
- Consistent troubleshooting workflow
- Human validation before final decision



Project Objectives

1 Analyze network symptoms

3 Detect common network faults

5 Map problems to OSI layers

7 Recommend next troubleshooting commands

9 Maintain human oversight

2 Process Cisco-style show-command outputs

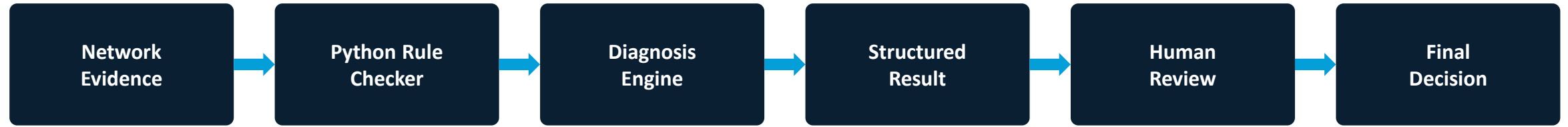
4 Identify likely root causes

6 Provide evidence-backed recommendations

8 Suggest remediation steps

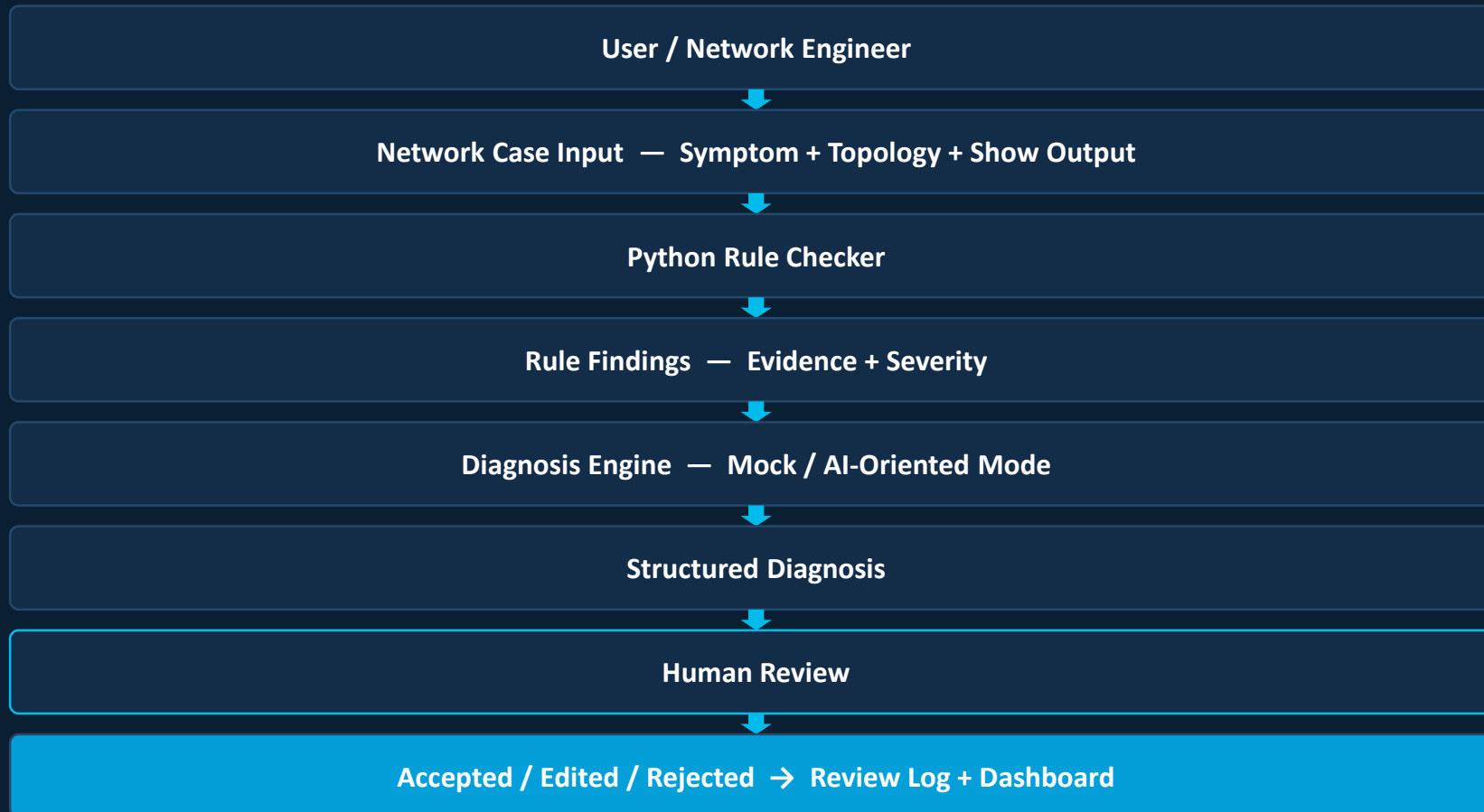
10 Foundation for future LLM integration

A Hybrid Troubleshooting Pipeline



- Deterministic network rules provide structured findings
- Diagnosis logic converts findings into a structured recommendation
- Human review validates the recommendation
- Review decisions can be recorded for future evaluation

End-to-End System Architecture



Technology Stack



Python

Core programming and troubleshooting logic



Streamlit

Web application interface



Pandas

Dataset processing



CSV

Data and review-log storage



Plotly / Streamlit Charts

Visualization and dashboard



OpenAI-Compatible Architecture

Planned / future LLM integration



Cisco Networking

Networking domain knowledge



VS Code + venv

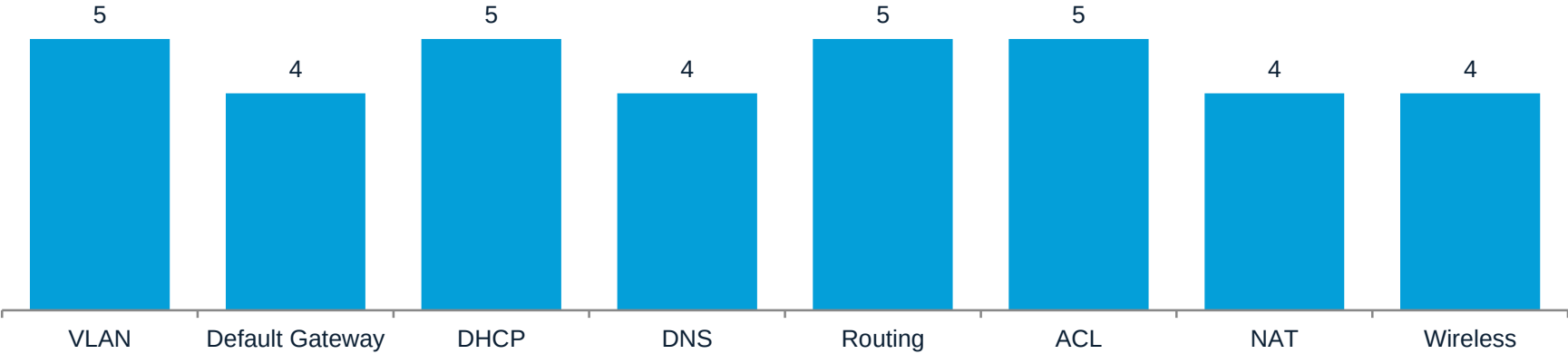
Development environment

Current prototype operates in mock/rule-assisted diagnosis mode due to API quota limitations.

Troubleshooting Dataset

36

TOTAL CASES

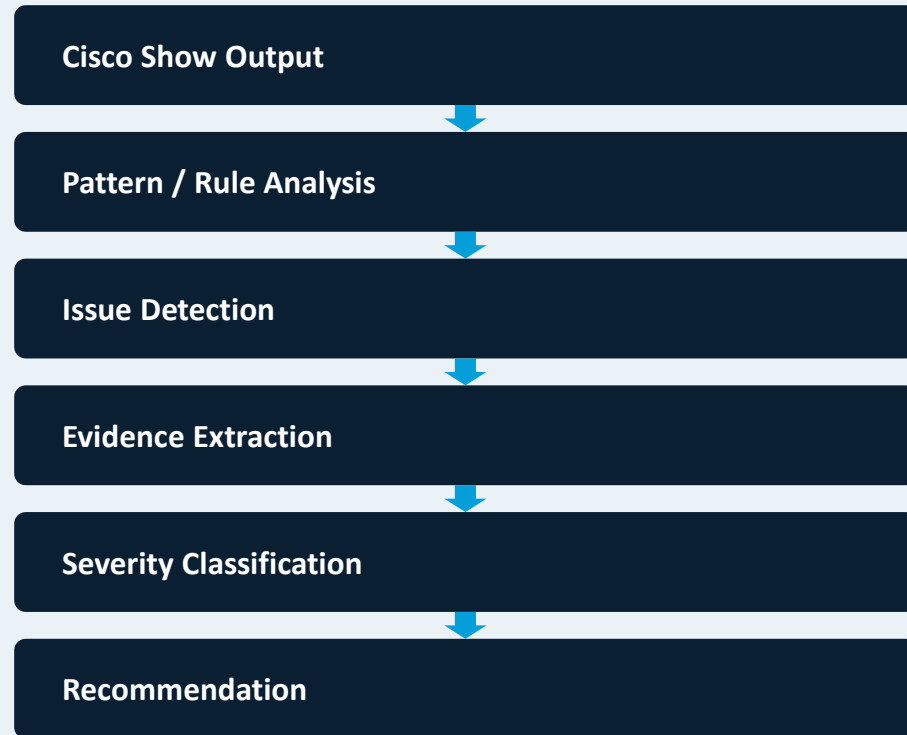


CASE FIELDS

Case ID	Symptom	Topology Note	Show Output
Expected Fault	OSI Layer	Concept	Severity

Structured, simulated Cisco-style lab troubleshooting scenarios — not production network data.

Python Rule Checker



EXAMPLES OF DETECTED PROBLEMS

- Interface administratively down
- Missing VLAN
- VLAN not allowed on trunk
- Gateway mismatch
- Missing route
- DHCP issue
- ACL issue
- NAT-related issue

Deterministic rules provide predictable and evidence-based findings before diagnosis.

Diagnosis Engine

INPUTS

Network Symptom

Topology

Cisco Show Output

Rule Findings



```
{ "root_cause": "...",  
  "osi_layer": "...",  
  "confidence": 0,  
  "evidence": [ ],  
  "next_command": "...",  
  "fix_steps": [ ] }
```



OUTPUTS

Root Cause

OSI Layer + Confidence

Evidence + Next Command

Fix Steps + Human Review

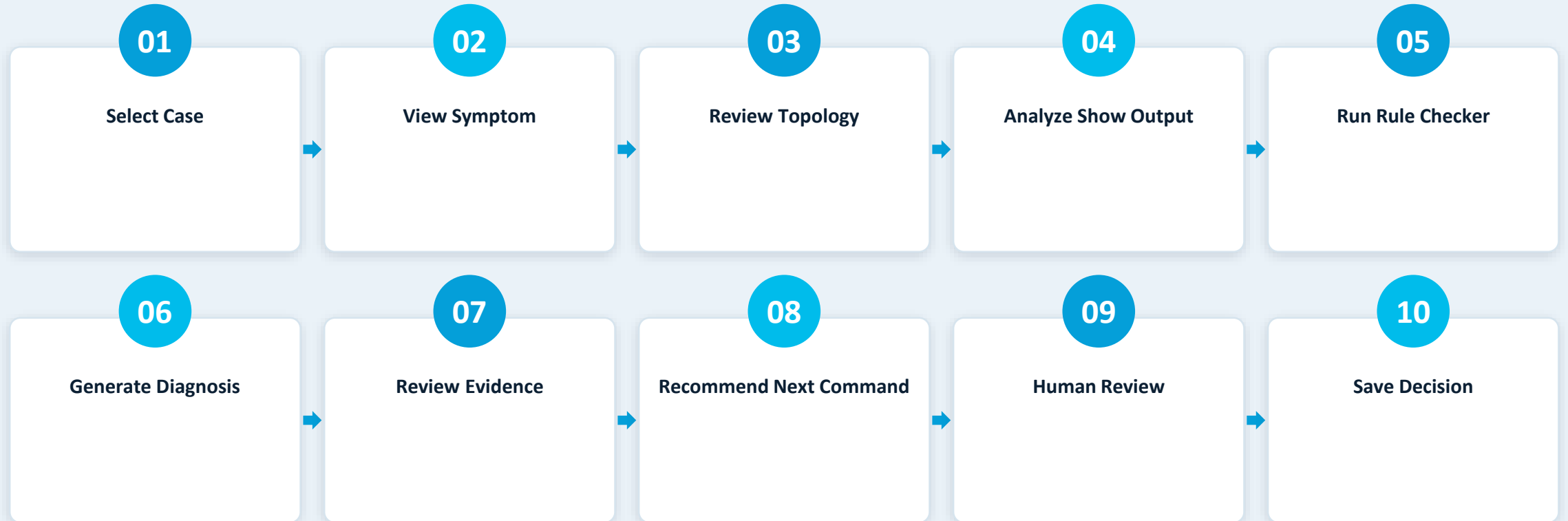
CURRENT

Mock / rule-assisted diagnosis

FUTURE

Real LLM-powered diagnosis

End-to-End Workflow



Streamlit Application

Illustrative interface concept — module layout as implemented in the Streamlit app

HOME

Project overview

Project purpose

Workflow explanation

DIAGNOSE NETWORK

Case selection

Symptom + topology

Show output

Rule findings

Diagnosis result

DIAGNOSIS RESULT

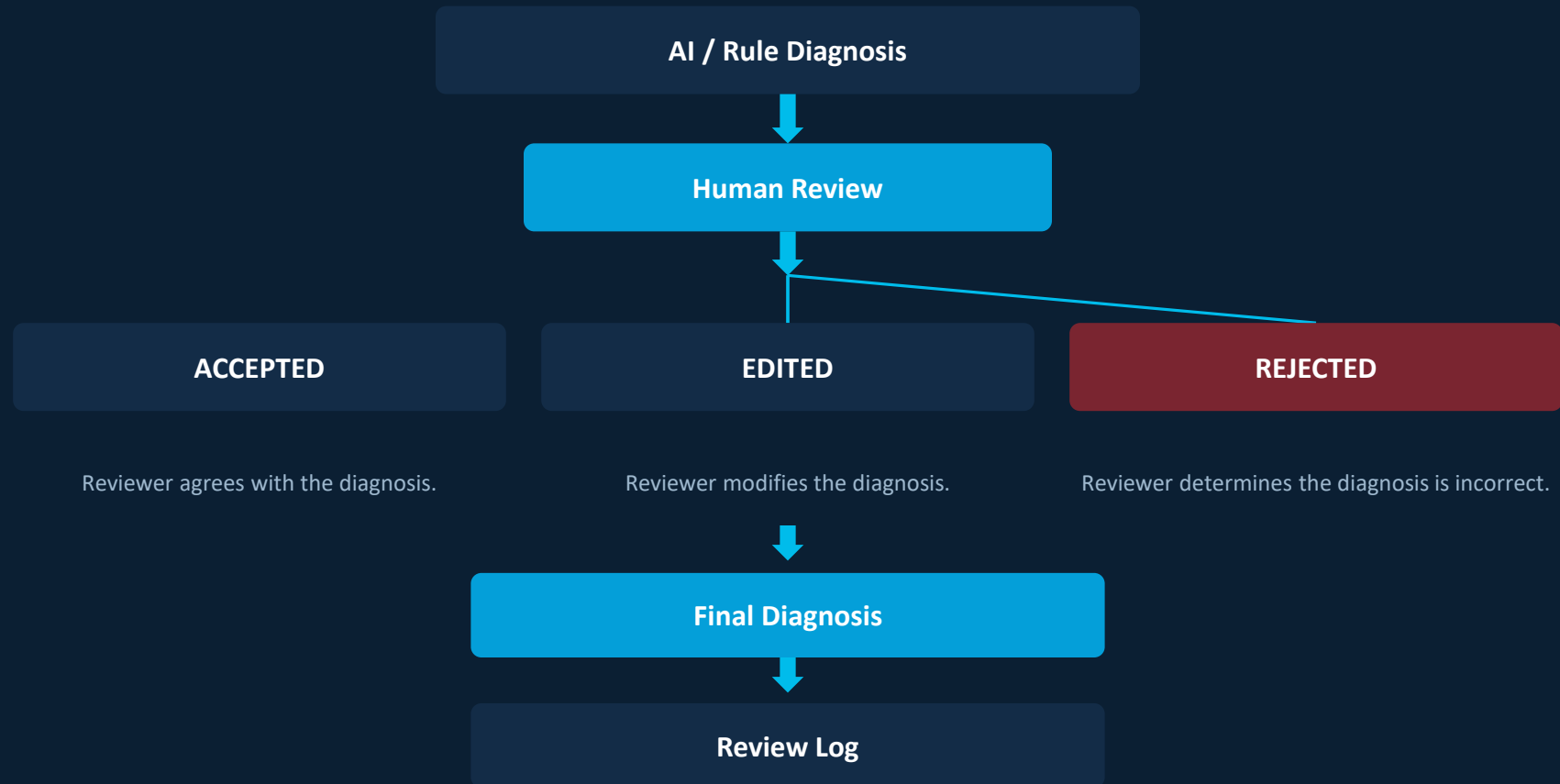
Evidence

Next command

Fix steps

Human review action

Human-in-the-Loop Review



"The system recommends; the human decides."

Dashboard & Analytics

IMPLEMENTED

- Total troubleshooting cases
- Cases by networking concept
- Cases by severity
- Dataset overview
- Review information where available

FUTURE

- Acceptance rate
- Edit rate
- Rejection rate
- AI-human agreement
- Confidence vs. correctness
- Most common network faults

Demonstration Scenario

DEMONSTRATION SCENARIO

SYMPTOM	PC cannot communicate with another device.
EVIDENCE	Interface is administratively down.
RULE FINDING	Interface status problem detected.
LIKELY ROOT CAUSE	Interface is administratively down.
OSI LAYER	Layer 1 — Physical
NEXT COMMAND	Check interface status and configuration.
RECOMMENDED FIX	Enable the interface (e.g., "no shutdown").
HUMAN REVIEW	Accepted

If this exact scenario is not present in the uploaded project, it is presented only as a demonstration — not experimental evidence.

TESTING AND VALIDATION

Testing & Validation

1

Dataset Testing

Validate dataset structure and required fields

2

Rule Checker Testing

Test common network fault conditions

3

Diagnosis Testing

Verify structured diagnosis output

4

Human Review Testing

Validate Accept / Edit / Reject workflow

5

UI Testing

Validate Streamlit application behavior

TEST CATEGORIES

VLAN

Gateway

DHCP

DNS

Routing

ACL

NAT

Wireless

Functional testing was used to validate application workflow and module behavior.

No claim of 100% accuracy is made. Numerical results are reported only where supported by the uploaded materials.

Challenges & Limitations

CHALLENGE	RESPONSE	STATUS
API quota limitation	Mock / rule-assisted mode	Architecture preserved for future LLM integration
Need for structured troubleshooting cases	36-case dataset	Implemented
Evidence-based diagnosis	Rule findings + structured outputs	Implemented
Human oversight	Accept / Edit / Reject workflow	Implemented

CURRENT LIMITATIONS

- Structured / simulated troubleshooting scenarios
- Simplified Cisco-style outputs
- No physical network-device integration
- Limited deterministic rule coverage
- CSV storage
- No large-scale AI accuracy evaluation

Responsible AI & Future Scope

RESPONSIBLE AI

- Human-in-the-loop
- Evidence-based reasoning
- No blind automation
- Review logging
- Explicit uncertainty
- Human correction
- Final decision remains with the reviewer

FUTURE SCOPE

- Real LLM API integration
- RAG with Cisco documentation
- Automated show-command parsing
- Real network-device integration
- SNMP monitoring
- Syslog analysis
- Real-time monitoring
- Predictive fault detection
- Database integration
- Advanced AI-human analytics

Prototype



AI-Assisted Assistant



Intelligent Network Operations Platform

CONCLUSION

Conclusion

- Developed a Python-based network troubleshooting prototype
- Covered 36 Cisco-style troubleshooting cases
- Implemented deterministic rule checking
- Designed an AI-oriented diagnosis workflow
- Implemented mandatory human review
- Built a Streamlit-based interface
- Incorporated responsible AI principles
- Created a foundation for future real LLM integration

“NetSage AI demonstrates how AI-assisted systems can support network troubleshooting while keeping humans responsible for the final decision.”

THANK YOU

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Questions?